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Malignant hyperthermia: Diagnosis and management of acute crisis

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INTRODUCTION

Malignant hyperthermia (MH) manifests clinically as a hypermetabolic crisis when an MH-susceptible (MHS) individual is exposed to a volatile anesthetic (eg, halothane, isoflurane, sevoflurane, desflurane) or succinylcholine [1-5].

This topic will discuss the incidence, pathophysiology, clinical manifestations, and acute management of MH. Susceptibility to MH and administration of anesthesia to MHS patients are discussed elsewhere. (See "Malignant hyperthermia susceptibility: Evaluation and management".)

INCIDENCE OF MH EVENTS

The incidence of MH events for a given population depends upon the prevalence of MH susceptibility and use of triggering anesthetics. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Prevalence'.)

MH episodes have been estimated to occur in the general population in 1:100,000 administered anesthetics [6]. This is probably an underestimate because unrecognized, mild, or atypical reactions occur due to variable penetrance of this autosomal dominant inherited trait.

An MH event does not necessarily occur every time an MH susceptible individual is exposed to an anesthetic triggering agent [7]. Approximately one-half of patients who develop acute MH have had one or two uneventful exposures to triggering agents [8,9]. In one case reported to the Malignant Hyperthermia Association of the United States (MHAUS) hotline and confirmed

by molecular genetic analysis, the patient had undergone approximately 30 general anesthetics prior to triggering MH. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Anesthetic history'.)

Epidemiology — MH occurs in all ethnic groups in all parts of the world. Reactions occur more frequently in males than females (2:1) [6,8,10]. Children under 19 years account for 45 to 52 percent of reported events [8,10].

PATHOPHYSIOLOGY

MH-susceptible (MHS) patients have genetic skeletal muscle receptor abnormalities that allow excessive myoplasmic calcium accumulation in the presence of certain anesthetic triggering agents. The specific mechanisms by which anesthetics interact with these abnormal receptors and trigger an MH crisis have not been defined, but appear to involve impaired magnesium inhibition of SR calcium release or leak and extracellular calcium entry [4,5,11,12].

During an episode of MH, the clinical manifestations occur as a result of calcium overload within the skeletal muscle cell that leads to sustained muscular contraction and breakdown (rhabdomyolysis), cellular hypermetabolism, anaerobic metabolism, acidosis, and their sequelae.

- **Normal muscle physiology** – Depolarization spreads throughout the muscle cell via the transverse tubule system, which activates dihydropyridine (DHP) receptors located within the t-tubule membrane. These receptors are coupled to ryanodine receptors (*RYR1*), which regulate the passage of calcium from the sarcoplasmic reticulum into the intracellular space [13,14]. Calcium combines with troponin to cross-link actin and myosin, resulting in muscle cell contraction. Reuptake of calcium by the sarco(endo)plasmic reticulum calcium ATPase (SERCA) leads to muscle cell relaxation.

- **Malignant hyperthermia pathophysiology** – Mutations encoding for abnormal RYR1 or DHP receptors have been found in a majority of MHS patients; exposure to triggering agents in these patients may lead to unregulated passage of calcium from the sarcoplasmic reticulum into the intracellular space combined with transient receptor potential canonical (TRPC) mediated transsarcolemmal Ca2+ influx, leading to an acute MH crisis [13-25]. This process is shown in a figure.

  The unregulated accumulation of myoplasmic calcium causes sustained muscle contraction. Accelerated levels of aerobic and anaerobic metabolism sustain the muscle for a time, but produce carbon dioxide and cellular acidosis, and deplete oxygen and adenosine triphosphate [26-28]. This causes the early signs of MH: hypercarbia, tachycardia, and in some cases, muscle rigidity. A change to anaerobic metabolism worsens acidosis with the production of lactate, resulting in a mixed respiratory/metabolic acidosis. Once energy

stores are depleted, rhabdomyolysis occurs and results in hyperkalemia and myoglobinuria. Clinically evident hyperkalemia (ie, electrocardiogram [ECG] changes) from rhabdomyolysis may occur early after succinylcholine or later with progression of the MH episode.

Over time, sustained muscle hypermetabolism generates more heat than the body can dissipate, aggravated by cutaneous vasoconstriction. Hyperthermia may occur early or may be delayed following the initial onset of symptoms. In some cases, core body temperature rises as much as 1°C every few minutes. Accelerated hyperthermia (above 41.5°C [106.7°F]) causes widespread vital organ dysfunction. These extremely high body temperatures are associated with the development of disseminated intravascular coagulation, a poor prognostic indicator and often terminal event [29]. (See 'Mortality' below.)

- The only known therapy for MH, dantrolene, binds to the RYR1 receptor to inhibit the release or leak of calcium from the sarcoplasmic reticulum and attenuates extracellular calcium entry [30]; this reverses the negative cascade of effects [31-33]. Dantrolene's effect requires the presence of calmodulin and sufficient magnesium [34].

  The mechanism whereby certain anesthetic agents trigger these events in MHS patients is unclear [11,12]. Prolonged RYR1 channel opening or channel leak has been demonstrated in experimental models of MH susceptibility [35]. Volatile anesthetics potentiate sarcoplasmic calcium release in patients with MHS [36]. Increased sarcolemmal influx of calcium via TRPC6 and TRPC3 cationic channels is as important as dysregulated SR calcium leak/release in MH pathophysiology. The threshold for SR calcium overload induced calcium release (SOICR) is lower with MH-causal RYR1 mutations, and further lowered by inhalation anesthetics. Increased mitochondrial calcium may produce reactive oxygen species which also increases SR calcium leak [37,38].

  The mechanism whereby succinylcholine administration can initiate acute MH has not been elucidated. Succinylcholine is an analog of acetylcholine and stimulates the motor endplate to initiate muscle depolarization, which may be prolonged in MHS patients. In MHS patients, succinylcholine and/or potent inhalational anesthetics may result in generalized muscle rigidity or masseter muscle rigidity. (See 'Clinical signs' below.)

MH TRIGGERS

The vast majority of cases of MH have occurred while the patient was receiving a volatile anesthetic agent (eg, halothane, enflurane, isoflurane, sevoflurane, desflurane) with or without administration of succinylcholine [8].

MH has been reported following administration of succinylcholine in the absence of an inhalation agent (eg, to facilitate endotracheal intubation or treat laryngospasm). The majority of such cases were reported in two publications. One was a series of 129 patients

who were biopsy- proven MH-susceptible (MHS), 20 of whom manifested their signs of MH with succinylcholine alone without coadministration of a volatile agent [39]. The second publication reported 14 cases of MH triggered by succinylcholine alone among 477 MH cases reported to the North American MH Registry (NAMHR) between 1987 and 2010 [40].

MH has been reported to occur during or when coming off cardiopulmonary bypass; the most common presenting signs include unexplained tachycardia, hypercarbia, and acidosis [41].

An acute and sometimes fatal syndrome with signs and symptoms similar to anesthetic-induced MH (ie, rigidity, rhabdomyolysis, hyperthermia) has been reported in MHS children and adults in the absence of exposure to triggering anesthetic agents. The patients who have developed this syndrome were usually, but not always, exposed to heat stress or exercise (see "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Exertional rhabdomyolysis'), and were either previously known to be MHS (ie, ryanodine receptor [*RYR1*] mutation positive) or subsequently shown to be on postmortem molecular genetic analysis [42-44].

PREPAREDNESS FOR MH

The supplies necessary to treat MH must be immediately available wherever general anesthetic triggering agents (ie, volatile gases and succinylcholine) are used (table 1) [45-47]. Inability to promptly start dantrolene when an MH crisis is identified, coupled with unavailability of dantrolene during transfer from the outpatient facility to a hospital and any subsequent delay in starting dantrolene, increases the risk of patient injury or death.

The need for dantrolene to be immediately available in facilities that do not use volatile anesthetics, and stock succinylcholine only for airway emergencies, is controversial. Recommendations and/or mandates from regulatory bodies and relevant organizations on this issue vary [47-50].

- The Malignant Hyperthermia Association of the United States (MHAUS) [48], Ontario, Massachusetts, and Tennessee require that facilities that stock succinylcholine have dantrolene available within 10 minutes of identifying a suspected MH crisis, and stock at least 700 mg (a 10 mg/kg dose for a 70 kg patient) on site. MHAUS takes the position that the availability of dantrolene allows clinicians to administer succinylcholine for life threatening airway obstruction without the fear that MH could occur in the absence of available treatment.

- In contrast, the American Association for Accreditation of Ambulatory Surgery Facilities [49], the State of Florida Board of Medicine, and a joint position statement from the Society for Ambulatory Anesthesia (SAMBA) and the American Society of Anesthesiologists [47] do not recommend or require dantrolene availability in facilities that do not use volatile

anesthetics, with or without stocking succinylcholine; they claim that patient safety and cost-effectiveness are better served by having succinylcholine on hand for treatment of laryngospasm, even in the absence of dantrolene for treatment of succinylcholine-induced MH, which has a significantly lower incidence than laryngospasm. SAMBA recommends that patients who are known to be MHS should not be cared for at facilities that stock succinylcholine but do not stock dantrolene.

- The European Malignant Hyperthermia Group guidelines recommend that dantrolene should be available wherever volatile anesthetics or succinylcholine are used, with the exception of prehospital emergency settings [50].

CLINICAL FEATURES

The sequence and timing of the clinical manifestations of MH vary from patient to patient, and most patients do not develop all signs of MH (table 2).

Sequence of clinical signs of malignant hyperthermia

Frequent initial clinical signs of MH
• Hypercarbia (elevated end-tidal CO_2) resistant to increases in minute ventilation
• Tachypnea (or breathing over the ventilator)
• Sinus tachycardia
Unusual initial clinical signs of MH
• Masseter spasm with or without administration of succinylcholine
• Generalized muscular rigidity
• Peaked T waves or other arrhythmia (including ventricular tachycardia or fibrillation) resulting from sudden acute hyperkalemia
• Hyperthermia (variable rates of rise)
• Sweating
Delayed clinical sequelae of MH
• Cola-colored urine (myoglobinuria) resulting from significant rhabdomyolysis
• Disseminated intravascular coagulation (from prolonged or severe hyperthermia)

Sequence of clinical signs of malignant hyperthermia

Table 2 - larger image below

Timing of presentation — The initial signs of MH may occur soon after induction with general anesthetic triggering agents (ie, volatile agents and/or succinylcholine) or any time during the maintenance phase of the anesthetic. Some cases will appear within minutes after turning off the inhalational anesthetic [51-55]. Some cases are indolent, and others are more accelerated in nature, such as the occurrence of sudden life-threatening arrhythmias from acidosis and hyperkalemia related to rhabdomyolysis.

- **Sequence of clinical signs** – The most reliable initial clinical sign heralding the development of acute MH is an **unexplained** increase in end-tidal carbon dioxide ($ETCO_2$), tachypnea, or breathing over the ventilator, despite minute ventilation that would usually maintain normocarbia. Other initial signs of acute MH may include sinus tachycardia, or masseter or generalized muscle rigidity, which may occur or persist despite paralysis with a non-depolarizing muscle relaxant [8,56]. On rare occasions, the presenting sign has been an unexpected electrocardiographic (ECG) indication of acute hyperkalemia, such as peaked T waves, disappearance of P waves, QRS widening, premature ventricular contractions, or even more rarely, ventricular tachycardia, ventricular fibrillation, or asystole.

Although it may occur at any point in the clinical course of MH, hyperthermia usually occurs after hypercarbia and tachycardia. While hyperthermia may be absent when the diagnosis is initially suspected, it may occur as early as 15 minutes after onset, and is one of the first three signs in the majority of patients. Malignant Hyperthermia Association of the United States (MHAUS) strongly recommends accurate core temperature monitoring to avoid delayed diagnosis and treatment of MH, which has occurred in the absence of temperature monitoring or with monitoring skin temperature.

Postoperatively, isolated rhabdomyolysis may occur in otherwise asymptomatic patients that have received triggering agents. These patients do not have classic clinical signs of MH, and it is unclear whether these episodes represent true MH crises or are due to ischemic injury or other heritable conditions that predispose to rhabdomyolysis [57-61].

- **Recrudescence** – In an analysis of MH cases reported to the North American MH Registry (NAMHR), recrudescence occurred in approximately 20 percent (63 of 308) of patients after successful treatment of an acute event [62]. Signs of recrudescence included tachycardia, increasing minute ventilation (doubling to tripling) to maintain $ETCO_2$, and increasing temperature. A time period of two hours or more after the initial MH event was used to define recrudescence. Temperature increase was defined as "an inappropriate temperature greater than 38.8°C in the perioperative period or an inappropriately rapid increase in temperature in the anesthesiologist's judgment." Recrudescence was more likely in patients with increased muscle mass and those who experienced a temperature increase during the initial episode. Patients with recrudescence were more likely to develop postoperative organ failure.

Clinical signs

Hypercarbia — The most consistent presenting manifestation of MH is hypercarbia, signaled by an increase in $ETCO_2$ that is resistant to increases in minute ventilation, and is not caused by hypoventilation, CO_2 rebreathing, or CO_2 absorption (eg, during laparoscopy) (see 'Anesthesia/surgery-related diagnoses' below). During general anesthesia without paralysis, patients may increase their rate or depth of spontaneous ventilation in response to hypercarbia.

Generalized muscle rigidity — Generalized muscle rigidity (ie, sustained contracture) in the presence of neuromuscular blockade is considered pathognomonic for MH, provided other confirmatory signs of hypermetabolism are also present. In a series of 255 patients, 40.8 percent developed generalized muscular rigidity [8].

Masseter muscle rigidity — Masseter muscle rigidity (MMR) in the context of MH is defined as jaw muscle tightness after administration of a triggering agent, usually succinylcholine, in the absence of temporomandibular joint dysfunction or myotonia. Severe MMR (ie, inability to manually open the patient's mouth) after succinylcholine may

indicate development of MH [56]. In a review of 41 cases referred for MH testing because of an episode of MMR, there were no clinical characteristics that could be used to differentiate patients who were MHS from those who were not MHS, as determined by genetic and contracture testing [63]. Furthermore, severity of MMR was not associated with likelihood of MHS.

Historically, MMR after administration of a triggering agent was thought to be an early sign of MH, and many of the patients tested for MH because of MMR history were positive by contracture test [64]. However, patients sent for testing were most likely those with the most severe MMR. Some increase in jaw tension is normal following succinylcholine administration, and very few patients with mild MMR develop MH [65,66]. Increased masseter muscle tone that is easily overcome with normal efforts at airway management following succinylcholine is of no particular concern, as long as it terminates within approximately one minute and is not associated with generalized rigidity.

Arrhythmias — Patients with MH frequently develop sinus tachycardia (table 3). They can also exhibit peaked T waves, QRS widening, or arrhythmias (eg, ventricular ectopy, tachycardia, fibrillation, or asystole) due to acute hyperkalemia. (See 'Laboratory findings' below and "Clinical manifestations of hyperkalemia in adults", section on 'Cardiac manifestations'.)

Hyperthermia — We agree with the recommendations from the MHAUS that core temperature (eg, nasopharyngeal, bladder, tympanic, or rectal) should be monitored for general anesthetics lasting more than 30 minutes [67]. Skin liquid crystal temperature indicators do not accurately trend with core temperature [8]. (See "Perioperative temperature management", section on 'Temperature monitoring'.)

Accurate temperature monitoring may allow earlier diagnosis and treatment of MH. An updated analysis of NAMHR reports from 2007 to 2012 found that the risk of death with an MH event was increased nearly 14-fold when no temperature monitoring was used, compared with core temperature monitoring, and nearly 10-fold when skin temperature was monitored compared with core temperature [68]. (See 'Mortality' below.)

Although hyperthermia may occur at any point in the clinical course of MH, it is often absent when the diagnosis is initially suspected. Hyperthermia can occur as early as 15 minutes after onset of MH, usually after hypercarbia and tachycardia appear. An analysis of reports of MH events to the NAMHR for 1987 to 2006 found that elevated or rapidly increasing temperature was one of the first signs noted in only 8.2 percent, and the only initial sign in 3.9 percent, but was among the first three signs of an MH event in over 60 percent of patients with a mean temperature of 39.1°C [8]. A temperature elevation or rapidly increasing temperature occurred in over 50 percent of events (table 3). Higher maximum temperatures correlated with a higher likelihood of all complications of MH

events. Further, the study found that skin temperature monitoring did not track well with core temperature monitoring when both methods of measurement were used.

There is a widespread misconception that acute MH may begin in the late postoperative period with hyperthermia as the presenting sign. Although severe postoperative hyperthermia (ie, T >39°C) is relatively uncommon, the MHAUS hotline receives a disproportionate share of calls from practitioners worried about the possibility of postoperative MH. The development of hyperthermia more than one hour after discontinuing the triggering anesthetic agent, especially when other signs of MH are absent, should prompt a serious consideration of other causes.

Myoglobinuria — Brownish-, cola-, or tea-colored urine indicates the presence of myoglobinuria, which is due to rhabdomyolysis. Diagnosis of rhabdomyolysis and management of myoglobinuria are discussed separately. (See "Rhabdomyolysis: Clinical manifestations and diagnosis" and "Prevention and treatment of heme pigment-induced acute kidney injury (including rhabdomyolysis)".)

A number of reports have described seemingly normal patients with postoperative rhabdomyolysis and myoglobinuria without any of the other classic signs of MH. MH contracture testing in these patients may be positive; however, it is unclear whether this is due to true MH susceptibility, obesity, or position-associated muscle ischemia, or another muscle disorder resulting in false-positive test results [57-60]. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Malignant hyperthermia susceptibility testing' and "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Muscle disorders with intraoperative rhabdomyolysis'.)

Laboratory findings — Laboratory findings that support the diagnosis of MH are shown in a table (table 4).

- **Mixed metabolic and respiratory acidosis** – Almost all patients with MH events develop respiratory acidosis, and some develop metabolic acidosis as well. In a series of 196 cases of MH reported to the NAMHR between 1987 and 2006 with arterial blood gas (ABG) measurements available, 99 percent of patients developed respiratory acidosis and 26 percent developed a metabolic acidosis (all but one also had respiratory acidosis) [8].

- **Hyperkalemia** – Hyperkalemia can occur rapidly due to muscle breakdown during an MH event, especially in muscular patients. It is infrequently the presenting sign of MH. Acute hyperkalemia may manifest on the ECG as peaked T waves, premature ventricular contractions, disappearance of P waves, QRS widening, or even ventricular tachycardia, ventricular fibrillation, or asystole.

- **Elevated creatine kinase and myoglobinuria** – Plasma creatine kinase (CK) levels peak approximately 14 hours after an acute MH episode. Peak CK levels depend upon the muscle

mass of the patient and severity of muscle breakdown; in some patients, levels may exceed 100,000 units/L.

- **Disseminated intravascular coagulation** – Death due to MH is often associated with the development of disseminated intravascular coagulation (DIC) and end-stage organ failure [8,69]. This is likely due to severe and prolonged hyperthermia, rhabdomyolysis, and acidosis [70]. Patients with ryanodine receptor [*RYR1*] mutations may have occult bleeding tendencies that may contribute to the development of DIC [71].

Pediatric presentation — Pediatric patients with acute MH present somewhat differently at different ages. In a retrospective analysis of data on patients under 18 years of age from the NAMHR, the most commonly observed physical findings in all children were sinus tachycardia (73.1 percent), hypercarbia (68.6 percent), and rapid temperature increase (48.5 percent) (table 5) [72]. The youngest children (age 0 to 24 months) were half as likely to have muscle rigidity, but had skin mottling more often than older children; they also had higher peak lactic acid, and lower peak CK levels. Children age 25 months to 12 years had lower maximum $ETCO_2$ and carbon dioxide tension ($PaCO_2$) compared with the youngest and oldest age groups, but were over three times more likely to have masseter spasm. Children age 12 to 18 years had higher peak potassium levels, higher maximum temperatures, were more likely to sweat, and took longer to reach their maximum $ETCO_2$ levels.

DIAGNOSIS

MH should be strongly suspected when the end-tidal carbon dioxide ($ETCO_2$) increases despite significantly increasing minute ventilation. The diagnosis is further supported by, but does not require, muscle rigidity (generalized or prolonged masseter muscle rigidity [MMR]) or an otherwise unexplained metabolic acidosis. During an acute event, diagnosis of MH is presumptive, based upon a presence of one or more of the typical clinical signs associated with MH, without another persuasive clinical explanation; more features increase the strength of the presumptive diagnosis (table 6 and table 2). (See 'Clinical signs' above.)

Clinical indicators of acute malignant hyperthermia

Hypercarbia*	Controlled ventilation:
	$ETCO_2$ >55 mmHg
	$PaCO_2$ >60 mmHg
	Spontaneous ventilation:
	$ETCO_2$ >60 mmHg
	$PaCO_2$ >65 mmHg
Muscle rigidity	Generalized muscular rigidity (sustained contracture) during anesthesia with triggering agents
	Severe masseter muscle spasm shortly following succinylcholine administration that does not abate within 1 minute
Arrhythmias	Tachycardia
	Ventricular arrhythmias (indicating acute hyperkalemia)
Acidosis	Primarily respiratory acidosis but may include varying degrees of metabolic acidosis
	Arterial pH <7.25
	Base deficit >8 mEq/L
Hyperthermia	Core temperature >38.8°C
	Rapid rise in temperature
Rhabdomyolysis	CK >20,000 international units after anesthesia with succinylcholine
	CK >10,000 international units after anesthesia without succinylcholine
	Cola-colored urine in perioperative period
	Urine myoglobin >60 mcg/L
	Serum myoglobin >170 mcg/L
	Serum potassium >6 meq/L (without renal failure)

Clinical indicators of acute malignant hyperthermia

Table 6 - larger image below

There is no confirmatory test for MH during an acute event. The diagnosis must be considered in all patients with clinical signs who have received triggering agents, regardless of family history or prior uneventful anesthetics. Over 90 percent of patients developing acute

MH episodes have negative family histories for MH, and over half have had uneventful general anesthetics in the past [8]. After an MH event, the patient or a family member sometimes remembers a family member with a similar diagnosis or complication.

Treatment must be initiated urgently, as soon as an MH crisis is suspected, often before other diagnoses in the differential can be definitively ruled out.

Laboratory studies are not required for presumptive diagnosis, though characteristic abnormalities support the diagnosis (table 4). (See 'Laboratory findings' above.)

Following an acute event, the likelihood that a clinical event represented a true MH episode can be estimated using the MH clinical grading scale (calculator 1) [56]. Definitive diagnosis can only be achieved through susceptibility testing. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Identification of malignant hyperthermia-susceptible patients' and "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Anesthetic history'.)

DIFFERENTIAL DIAGNOSIS

The differential diagnosis for MH is wide, as a number of conditions commonly present in the perioperative period with the various clinical features that can occur with MH (eg, hypercarbia, tachycardia, arrhythmia, hyperthermia). Although treatment for MH may have been initiated, it is important to continue to consider other causes. (See 'Clinical Features' above.)

Anesthesia/surgery-related diagnoses — Several relatively common intraoperative conditions in the differential diagnosis of MH can cause hypertension, tachycardia, tachypnea, hypercarbia, and/or fever. None of these conditions is associated with muscle rigidity, masseter spasm, rhabdomyolysis, hyperkalemia, or metabolic acidosis.

- **Insufficient anesthesia/analgesia** – Patients with insufficient anesthesia/analgesia can have tachycardia, hypertension, and tachypnea in a spontaneously-breathing patient. In contrast with MH, tachypnea in due to light anesthesia would usually be associated with reduced end-tidal carbon dioxide ($ETCO_2$). In the absence of neuromuscular blockade involuntary muscle activity, such as gross tremor with increased muscle tone, may also occur with insufficient anesthesia.

- **Hypoventilation or CO_2 rebreathing** – Patients with insufficient ventilation have hypercarbia, respiratory acidosis, and may have reflex tachycardia and hypertension. Insufficient ventilation can be the result of inappropriate ventilator settings, or equipment problems (eg, malfunction of unidirectional valves on the anesthesia breathing circuit, kinked endotracheal tube, exhausted CO_2 absorber, increased dead space). Presence of

significant inspired CO_2 on the anesthesia gas analyzer does not occur at the onset of an MH episode, before the CO_2 absorbent is exhausted, and usually indicates an equipment problem.

- **Increased CO_2 absorption during laparoscopy or GI endoscopy** – Hypercarbia resistant to increases in minute ventilation may be due to continuous CO_2 absorption during laparoscopy or less commonly during GI endoscopy. The presence of subcutaneous emphysema, or known insufflation of CO_2 into tissues, makes this a likely explanation. Tachycardia and hypertension are also common during laparoscopy. (See "Anesthesia for laparoscopic and abdominal robotic surgery in adults", section on 'Carbon dioxide insufflation'.)

- **Fever** – Fever alone, no matter how high, is not a useful indicator of acute MH. This may occur as a result of an inflammatory or infectious process or iatrogenic overwarming. Postoperative fever is relatively common, and in the absence of other signs and symptoms of MH, alternate diagnoses should be sought.

Others — Other conditions in the differential diagnosis can be considered, depending on the clinical circumstances:

- Sepsis

- Anaphylaxis (see "Anaphylaxis: Acute diagnosis")

- Anesthesia induced rhabdomyolysis (see "Anesthesia for children with myopathy and for children who undergo muscle biopsy", section on 'Anesthesia-induced rhabdomyolysis and myopathy')

- Thyrotoxicosis or thyroid storm (see "Thyroid storm")

- Transfusion reaction (see "Immunologic transfusion reactions")

- Drugs of abuse (ie, cocaine, ecstasy [MDMA], methamphetamine, "bath salts") (see "Cocaine: Acute intoxication" and "MDMA (ecstasy) intoxication" and "Methamphetamine: Acute intoxication")

- Serotonin syndrome (see "Serotonin syndrome (serotonin toxicity)")

- Pheochromocytoma (see "Clinical presentation and diagnosis of pheochromocytoma")

- Neuroleptic malignant syndrome (see "Neuroleptic malignant syndrome")

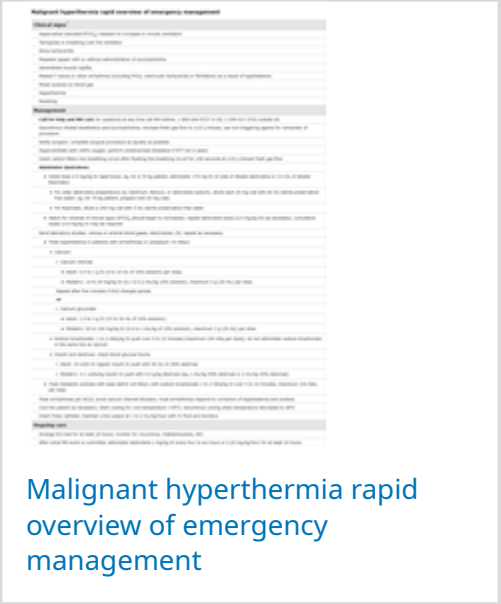
- Following hypoxic brain injury or hypothalamic injury (see "Pathophysiology and treatment of fever in adults", section on 'Fever')

- Baclofen withdrawal [73]

ACUTE MANAGEMENT OF SUSPECTED MH

Treatment for MH must be initiated immediately when the following occur in the absence of a persuasive alternative diagnosis (table 7):

- Rising end-tidal carbon dioxide ($ETCO_2$) despite compensatory increase in minute ventilation
- One or more of the other clinical signs of MH (eg, hyperthermia, muscle rigidity [generalized or prolonged masseter muscle rigidity (MMR)]), increased heart rate/tachycardia, or electrocardiogram (ECG) changes consistent with hyperkalemia. (See 'Clinical signs' above.)



Malignant hyperthermia rapid overview of emergency management

Table 7 - larger image below

Initial steps

- **Call for help and the MH cart** – Additional personnel should be mobilized, as care of a patient with MH is labor-intensive (table 7). The MH treatment cart should be brought into the immediate area (table 1).

  Assistance in diagnosing and managing an MH crisis is available from the Malignant Hyperthermia Association of the United States (MHAUS) hotline at 1-800-644-9737 in the United States (00+1+209-417-3722 outside the United States). An acute management protocol can be found on the MHAUS website, at www.mhaus.org. Other cognitive aids include the MH section in the Stanford Emergency Manual for Perioperative Crises and the MHApp endorsed by the European MH Group and MHAUS.

- **Notify the surgeon** – Halt the surgical procedure as soon as possible.
- **Optimize oxygenation and ventilation** – Increase inspired oxygen to 100 percent and maximize fresh gas flow. Increase mechanical ventilation rate and/or tidal volume (eg, to threefold normal minute ventilation) to maximize ventilation and reduce the $ETCO_2$. If the patient is not intubated, place an endotracheal tube using intravenous anesthetics or sedative-hypnotics and non-depolarizing muscle relaxants if paralysis is required, and institute mechanical ventilation.
- **Discontinue triggering agents** – Immediately discontinue volatile anesthetic agents and increase fresh gas flow to ≥10 L/minute to enhance elimination of anesthetic gas. Replace the disposable breathing circuit and reservoir bag, and if available, insert activated charcoal filters into the inspiratory and expiratory limbs of the anesthesia breathing circuit (picture 1). It is not necessary to change the anesthesia machine. If surgery must be

continued, maintain general anesthesia with intravenous non-triggering agents, typically propofol with opioids as needed, though other intravenous hypnotics, analgesics, or anesthetics (eg, midazolam or ketamine) may be administered.

Administer dantrolene — Administer dantrolene as soon as the drug is reconstituted. Since this is labor intensive with the older formulations of dantrolene, extra personnel should be summoned to assist with mixing. (See 'Dantrolene' below.)

- Administer a loading dose of 2.5 mg/kg IV (intravenous; actual body weight) rapidly through a large bore IV if possible. Do not delay IV administration if limited to small gauge IV access.
  - For older formulations of dantrolene (Dantrium, Renovo, generic dantrolene sodium), dilute each 20 mg vial with 60 mL sterile water for injection to a concentration of 0.33 mg/mL. For a 70 kg patient, 175 mg (9 vials) will be required.
  - For Ryanodex, dilute the 250 mg vial with 5 mL sterile water for injection to a concentration of 50 mg/mL. For a 70 kg patient, 175 mg (3.5 mL) will be required.
- Administer subsequent doses of 2.5 mg/kg IV every five minutes until the signs of acute MH begin to abate (eg, $ETCO_2$ <50 mmHg (6.7 kPa) and able to reduce minute ventilation without recurrence of hypercarbia, resolution of rigidity if present, and temperature no longer increasing). Up to 10 mg/kg IV or more may rarely be required in some patients, especially muscular males with generalized rigidity. (See 'Efficacy of dantrolene' below.)

Laboratory monitoring and treatment of abnormalities — An arterial catheter should usually be placed to facilitate measurement of electrolytes, blood gases for acid/base status, creatine kinase (CK), and depending on the patient's condition, serum or urine myoglobin, coagulation parameters, including fibrinogen and platelet count (table 4). Arterial or (non-tourniquet) venous blood gases should be measured initially and as needed until pH and potassium levels trend towards normal values [74].

- **Hyperkalemia** – Treat hyperkalemia (ie, with calcium chloride, insulin-glucose, sodium bicarbonate) to prevent the development of life-threatening arrhythmias or cardiac arrest. In patients who have received insulin, inhaled or nebulized albuterol may also be given at a dose fourfold higher than used for bronchospasm. We treat hyperkalemia in patients with abnormal ECG waveforms (eg, peaked T waves, disappearance of P waves, QRS widening, ventricular arrhythmias), or patients with potassium of ≥6 mEq/L even in the absence of ECG abnormalities. (See "Treatment and prevention of hyperkalemia in adults" and "Hyperkalemia in children: Management".)
- **Metabolic acidosis** – Consider administering sodium bicarbonate (1 to 2 mEq/kg IV) for base deficit greater than 8 mEq/L. Each 50 mEq sodium bicarbonate results in production of

1 L carbon dioxide. Therefore, bicarbonate should be administered over several minutes while maintaining high minute ventilation.

Cardiovascular support

- Treat cardiac arrhythmias as per advanced cardiac life support (see "Advanced cardiac life support (ACLS) in adults"). Arrhythmias usually respond to the treatment of acidosis, hyperkalemia, and hyperthermia.
- **Avoid verapamil or diltiazem** – Use of verapamil or diltiazem to treat arrhythmias or hypertension is **contraindicated** during an MH crisis because of the possibility that it can worsen hyperkalemia, myocardial depression, and hypotension when co-administered with dantrolene [75]; however, dantrolene should never be withheld as treatment for MH in patients receiving preoperative maintenance therapy with calcium channel blockers.

Cool as necessary — Institute cooling for patients with core temperature >39°C, and discontinue cooling when temperature decreases to 38°C. Uncover the patient and rapidly administer cool or cold isotonic crystalloid (20 to 30 mL/kg IV) for patients without signs of congestive heart failure. In larger pediatric patients and adults, place ice packs at the neck, groins, or axillae; multiple ice packs or cooling with a circulating water mattress can also be used. If further cooling is necessary, consider cold saline lavage of open body cavities or via peritoneal catheter [76]. (See "Nonexertional (classic) heat stroke in adults", section on 'Cooling measures and temperature monitoring'.)

Monitor urine output — Insert a bladder catheter to monitor urine color and volume. A urine dipstick positive for heme (without red blood cells on microscopy) indicates myoglobinuria or hemoglobinuria. Maintain urine output at 1 to 2 mL/kg/hour. (See "Prevention and treatment of heme pigment-induced acute kidney injury (including rhabdomyolysis)", section on 'Prevention'.)

Refractory MH — Extracorporeal membrane oxygenation (ECMO) may be considered as a last resort for patients with persistent cardiac arrest unresponsive to the treatments in the MH protocol. Successful use of ECMO has been reported in two such cases [77,78]. (See "Extracorporeal life support in adults in the intensive care unit: Overview".)

DANTROLENE

We recommend administration of dantrolene as soon as MH is suspected, as dantrolene is the only known antidote for MH.

Timing of administration — An analysis of cases reported to the North American Malignant Hyperthermia Registry (NAMHR) indicate that the likelihood of an MH complication increased 1.6 times for every 30 minute delay between the first MH sign and the first dantrolene dose

[8]. Similarly, in a review of MH events in 373 patients referred to the Malignant Hyperthermia Unit in Canada, the time between onset of the first clinical sign and dantrolene administration was longer in patients who experienced complications, mostly renal dysfunction, compared with those who did not (23.5 versus 15.0 minutes, P = 0.005), and for each 10 minute delay in administration of dantrolene, complications increased substantially [39]. All patients who received dantrolene more than 50 minutes after the first clinical MH sign experienced complications.

Efficacy of dantrolene — Administration of an initial bolus dose of 2.5 mg/kg IV dantrolene will achieve therapeutic blood levels [79] and has been associated with resolution of clinical and laboratory signs of MH in a multicenter observational study [33]. The end-tidal carbon dioxide ($ETCO_2$) will usually decrease as the dantrolene takes effect; in most cases, dantrolene reverses the acute hypermetabolic process within minutes. The need to use doses beyond 10 mg/kg is uncommon, and the clinician should question the diagnosis if a response is not seen after a total dose of 10 mg/kg. However, some patients, especially muscular males with generalized rigidity, may require intravenous (IV) dantrolene doses ≥10 mg/kg during an acute event.

Since the introduction of Ryanodex into clinical practice, reports to the Malignant Hyperthermia Association of the United States (MHAUS) hotline of its use to treat acute MH appear to indicate that it has efficacy and side effects (eg, thrombophlebitis, weakness, nausea) comparable to the older generic versions of dantrolene.

Redosing dantrolene during acute MH — In most cases of MH, signs of hypermetabolism (eg, hypercarbia) will begin to decrease shortly after the initial bolus of dantrolene. Some patients will require additional doses for signs of hypermetabolism to completely abate. In a review of 286 cases of MH reported to the NAMHR between 1987 and 2006, the mean initial dose of dantrolene (older formulation) was 2.4 mg/kg, whereas the mean total dose was 5.9 mg/kg [8]. Definitive recommendations on re-dosing dantrolene following the initial bolus do not exist, and the need to redose should be individualized based on the patient response [74]. MHAUS and the European Malignant Hyperthermia Group (EMHG) recommend redosing dantrolene (2.5 mg/kg following the initial 2.5 mg/kg bolus) as frequently as needed until the patient responds with a decrease in $ETCO_2$, decreased muscle rigidity, and/or lowered heart rate [80,81].

The newer, hyperconcentrated formulation of dantrolene (Ryanodex) achieves dantrolene blood levels faster than the older formulation, but data on the speed and efficacy of treatment and on the need to redose are lacking.

Available preparations — In the United States, there are two types of dantrolene preparations. The older conventional, now generic, formulation is supplied as a lyophilized powder in a 20 mg vial, containing sodium hydroxide to maintain pH of 9 to 10 and 3 g of

mannitol, which can cause fluid volume and electrolyte complications (see "Complications of mannitol therapy"). Each 20 mg vial requires mixing with 60 mL of sterile water for injection [82]. It is important to summon additional personnel to assist with drug preparation and administration; the initial bolus of dantrolene in a 70 kg patient will require the mixing and administration of nine vials of the conventional preparation, at a time when multiple other interventions are required.

A newer dantrolene formulation (Ryanodex), which is dissolved rapidly, became available for clinical use in 2014. It is supplied in 250 mg vials, reconstituted with only 5 mL of sterile water, and warming is not needed. Because it is hyperconcentrated, blood concentrations will be achieved faster in patients with acute MH, with less of a sterile water volume load than the older form. Ryanodex contains a small, clinically irrelevant amount of mannitol (125 mg per vial).

The ease of use, shortened preparation time, lack of mannitol, and faster effective blood concentrations achieved with use of Ryanodex makes it the obvious choice for treating acute MH in larger children or adults or in settings with limited personnel, (eg, freestanding ambulatory surgery center). However, at present, there are no data comparing outcomes between MH patients treated with Ryanodex and the older generic formulations of dantrolene, nor is there an economic analysis that would justify its use, given the increased cost and shorter shelf-life of Ryanodex (currently 33 months versus 36 months for older generic versions of dantrolene).

Adverse effects — Dantrolene has no effect on cardiac or smooth muscle. Its most common local adverse reaction is venous irritation or thrombosis at the site of administration due to its high pH; side effects include nausea, malaise, lightheadedness, and mild to moderate muscle weakness [83]. Respiratory muscle weakness may occur when larger doses are used, especially in patients who are debilitated. Hepatotoxicity has been associated with long-term oral administration [84], but has not been reported after administration for MH.

Ryanodex appears to exhibit a favorable clinical and side effect profile that does not differ from existing dantrolene formulations [85].

ONGOING CARE

Following completion of the surgical procedure, the patient should be transferred to an intensive care unit for ventilatory support as needed with metabolic and hemodynamic monitoring for at least 24 hours.

Malignant Hyperthermia Association of the United States (MHAUS) recommends continuing maintenance doses of dantrolene (1 mg/kg intravenous [IV] bolus every four to six hours or 0.25 mg/kg/hour IV continuous infusion) for at least 24 hours after the last observed

sign of acute MH [62,76,86]. Recrudescence of MH occurs in approximately 20 percent of patients after initial treatment, at a mean of 13 hours (standard deviation, 13 hours) after the initial reaction [62]. If recurrent signs appear in spite of ongoing treatment, additional dantrolene boluses may be required.

Dantrolene can be stopped, or the interval between doses increased to every 8 or 12 hours if all of the following criteria are met:

- Metabolic stability for 24 hours
- Core temp is less than 38°C
- Creatine kinase (CK) is not increasing
- No evidence of myoglobinuria
- Muscle is no longer rigid

Patients should be monitored for disseminated intravascular coagulation (DIC) after an MH event. DIC was reported in approximately 7 percent of episodes of MH in one review, and was associated with higher maximum temperatures than in patients who did not develop DIC (40.3°C versus 39.0°C) [8]. (See "Evaluation and management of disseminated intravascular coagulation (DIC) in adults".)

Patients who develop rhabdomyolysis should be monitored for compartment syndrome, especially in patients with DIC. Muscle compartment release (ie, four compartment fasciotomy) may be required. (See "Acute compartment syndrome of the extremities".)

Prevention and treatment of acute kidney injury in patients with rhabdomyolysis is discussed separately. (See "Prevention and treatment of heme pigment-induced acute kidney injury (including rhabdomyolysis)", section on 'Prevention'.)

An increase in hemoglobin or hematocrit may reflect hypovolemia associated with fluid sequestration in injured muscle or mannitol-induced osmotic diuresis with traditional formulations of dantrolene.

MORTALITY

Estimates of mortality from MH have come from data derived from the National Inpatient Sample [87], and from reports submitted to the North American Malignant Hyperthermia Registry (NAMHR) [68,69]. Mortality from MH has declined significantly with the routine use of end-tidal carbon dioxide ($ETCO_2$) monitoring and availability of dantrolene, and is reported to be between 6 and 10 percent [68,87].

Core temperature monitoring may reduce mortality from an MH event. In reported cases of MH with simultaneous skin and core temperature monitoring, skin temperature did not accurately trend with core temperature [8]. Continuous core temperature monitoring allows

more rapid diagnosis of an MH event, more rapid treatment, and reductions in peak temperature and duration of hyperthermia, compared with no monitoring or skin temperature monitoring.

In an analysis of suspected MH reports submitted to Malignant Hyperthermia Association of the United States (MHAUS) between 1987 and 2006, there were 84 cases of likely MH, and 8 patients who died [68]. Patients whose temperature was not monitored were at least twice as likely to die, and patients with skin temperature monitoring were at least 1.5 times as likely to die, compared with those who had core temperature monitoring, based on the confidence interval's lower limit [68]. All deaths occurred in patients with a peak temperature of 38.9°C or higher.

Other factors that increase risk for cardiac arrest and death with MH include advanced age, comorbidities, heavy muscular build (eg, young males), and the development of disseminated intravascular coagulation (DIC) [87].

COUNSELING AFTER ACUTE MH

Following recovery from a suspected acute MH event, testing for MH susceptibility should be offered to the patient and family members. Evaluation and management for MH susceptibility is discussed in detail separately. (See "Malignant hyperthermia susceptibility: Evaluation and management".)

We counsel patients that until definitive testing for MH susceptibility (MHS) is complete, they should:

- Not have anesthesia with triggering agents.
- Obtain conspicuous identification (eg, MedicAlert bracelet) that they are MH susceptible, to inform medical providers in an emergency.
- Inform family members of the possible MH episode, as MHS is a genetic condition. Blood relatives are at risk and may also need to be evaluated. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Testing for family members'.)

In addition, if during intense or prolonged exercise the individual starts to feel ill, ataxic, confused or develops severe muscle cramps or rigidity, they should stop the activity and seek immediate medical attention. (See 'Non-anesthesia-related MH like events' below and "Nonexertional (classic) heat stroke in adults".)

MHS patients are encouraged to learn as much as possible about the nature of their condition and should be directed to the appropriate educational resources. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Counseling' and

"Malignant hyperthermia susceptibility: Evaluation and management", section on 'Malignant hyperthermia resources'.)

NON-ANESTHESIA-RELATED MH LIKE EVENTS

MH susceptibility (and ryanodine receptor [RYR1] myopathy) should be in the differential diagnosis for patients who exhibit unexplained stress-induced fever, muscle cramping, rigidity, myoglobinuria, or other characteristics of MH, unrelated to exposure to anesthesia. There is a rare subset of MH susceptible (MHS) children and adults who have developed what some authors call "non-anesthetic," or "awake," MH. Patients who have developed this syndrome were usually, but not always, exposed to heat stress, including febrile illness, or exercise in the absence of triggering anesthetic agents. In most cases, symptoms have abated either spontaneously or with self-administration of oral dantrolene, but several cases have rapidly proceeded to accelerated hyperthermia, hyperkalemia, and death [43,44,88]. Very little is known about the use of oral dantrolene for these patients, but in several reports doses as low as 25 mg taken at the time of the episode have alleviated symptoms [88,89]. (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Non-anesthesia-related MH-like episodes'.)

For patients who present with high fever, hyperkalemia, creatine kinase elevation, and muscle rigidity, even in the absence of exposure to anesthetics, treatment with dantrolene, which would not ordinarily be administered for heat stroke or exertional heat illness, should be considered. (See 'Dantrolene' above.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Malignant hyperthermia".)

SUMMARY AND RECOMMENDATIONS

- **Pathophysiology**
  - Malignant hyperthermia (MH) is an autosomal dominant disorder that may present with a hypermetabolic crisis when susceptible individuals are exposed to volatile anesthetics or succinylcholine. (See 'Introduction' above and 'MH triggers' above.)
  - MH-susceptible (MHS) individuals have skeletal muscle receptor abnormalities, most often in the ryanodine receptor (*RYR1*), which allow excessive intracellular calcium to

accumulate in response to triggering agents. This triggers intracellular events leading to skeletal muscle hypermetabolism. (See 'Pathophysiology' above.)

- **Preparedness for MH**
  - The supplies necessary to treat MH must be immediately available wherever triggering agents may be used (table 1). (See 'Preparedness for MH' above.)
  - Core temperature should be monitored during anesthetics lasting more than 30 minutes. (See 'Hyperthermia' above and 'Mortality' above.)
- **Clinical features and diagnosis**
  - Although the initial clinical signs of MH typically occur within one hour of anesthesia induction, the onset of MH can occur any time during the administration of triggering agents. The onset of MH in the postoperative period is extremely rare and does not generally manifest solely as temperature elevation. (See 'Clinical Features' above.)
  - The sequence and timing of the clinical manifestations of MH vary, and most patients do not develop all signs of MH. The diagnosis is based upon clinical signs (eg, hypercapnia, tachycardia, muscle rigidity, rhabdomyolysis, hyperthermia, and arrhythmia) and associated laboratory abnormalities (eg, respiratory and possibly metabolic acidosis, hyperkalemia, elevated creatine kinase, serum and urine myoglobin) (table 4). (See 'Clinical Features' above.)
  - The most reliable sign is unexplained hypercapnia (ie, increased end-tidal carbon dioxide [$ETCO_2$]) that is resistant to increasing the patient's minute ventilation (table 6). Tachycardia is another common early sign. Hyperthermia may occur as early as 15 minutes after onset of MH, and may be absent when the diagnosis is initially suspected (table 2). (See 'Clinical signs' above.)
  - The differential diagnosis for MH is wide, and should be considered. However, treatment for MH must be initiated urgently, as soon as the diagnosis of MH is considered reasonable, often before other diagnoses in the differential can be definitively ruled out. (See 'Differential diagnosis' above.)
- **Management** – As soon as the presumptive diagnosis of MH is made, call for assistance and the MH treatment cart (table 7) and initiate the following (see 'Acute management of suspected MH' above):
  - Complete the surgical procedure as soon as possible.
  - Discontinue volatile anesthetics, turn to 100 percent oxygen at ≥10 L/minute; replace the disposable breathing circuit and, if available, add charcoal filters to the inspiratory and

expiratory limbs. (picture 1). If necessary, continue anesthesia with non-triggering agents (eg, propofol). (See "Malignant hyperthermia susceptibility: Evaluation and management", section on 'Safe anesthetic agents'.)

- Hyperventilate and intubate if endotracheal tube is not already in place using nondepolarizing neuromuscular blocking agents, and institute mechanical ventilation.
- We recommend immediate administration of dantrolene (**Grade 1A**). The initial dose is 2.5 mg/kg intravenous (IV), to be given rapidly. The $ETCO_2$ typically normalizes within minutes; subsequent bolus doses of 2.5 mg/kg IV every five minutes (up to 10 mg/kg) may be needed if signs of MH have not abated. (See 'Dantrolene' above.)
- Assess for and treat hyperkalemia. (See "Treatment and prevention of hyperkalemia in adults".)
- Monitor blood gases, core temperature, creatine kinase (CK), urine output, urine color, electrolytes, coagulation parameters, and treat abnormalities as needed.
- Treat cardiac arrhythmias, which are usually responsive to correction of acidosis and hyperkalemia. Use advanced cardiac life support protocols. (See "Advanced cardiac life support (ACLS) in adults".)

- **Ongoing care**
  - After an MH event, patients should have supportive care in an intensive care unit for at least 24 hours and be closely monitored for recurrence. Continue dantrolene for 24 to 48 hours. (See 'Ongoing care' above.)
  - Following an acute event, until MH susceptibility is subsequently excluded, the patient should receive only non-triggering anesthetics, should limit exposure to excessive heat and humidity, and should inform family members of the diagnosis. (See 'Diagnosis' above and 'Counseling after acute MH' above.)
- **Non-anesthesia-related MH** – MH like events unrelated to anesthesia have rarely occurred in MHS patients. For patients who present with high fever, hyperkalemia, CK elevation and muscle rigidity, treatment with dantrolene should be considered, even in the absence of exposure to anesthetics. (See 'Non-anesthesia-related MH like events' above.)

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Topic 401 Version 56.0

GRAPHICS

Table 1: **Suggested supplies and equipment for the malignant hyperthermia treatment cart**

	Quantity per cart*
Drugs	
Dantrolene:	
▪ Dantrium, Revonto: 20 mg vials (lyophilized powder)	36 vials (enough supply to prepare maximum recommended cumulative dose [ie, 10 mg/kg] for one 72 kg patient)
OR	**OR**
▪ Ryanodex: 250 mg vials (lyophilized powder)	3 vials (enough supply to prepare maximum recommended cumulative dose [ie, 10 mg/kg] for one 75 kg patient)
Sterile water for injection (**non**-bacteriostatic):	
▪ 100 mL vials for reconstitution of Dantrium or Revonto	36 vials
OR	**OR**
▪ 5 mL vials for reconstitution of Ryanodex	3 vials
Sodium bicarbonate 8.4% (also labeled 1 mEq/mL or 1 mmol/mL), 50 mL¶	5 prefilled syringes or vials
Dextrose 50%, 50 mL¶	2 prefilled syringes or vials
Calcium chloride 10%, 10 mL¶	2 vials
Lidocaine 2%, 100 mg/5 mL **OR** 1%, 100 mg/10 mL¶	4 prefilled syringes or vials
Refrigerated drugs and solutions	
Insulin regular 100 units/mL, 10 mL	1 vial
0.9% normal saline, 1000 mL for IV cooling	4 bags and sterile pour bottles
Cold packs	8
General equipment	
Syringes, 60 mL to dilute dantrolene (unnecessary if using Ryanodex)¶	5 syringes
IV catheters (for IV access and arterial line): 16G, 18G, 20G 2-inch, 22G 1-inch, 24G ¾-inch¶	4 each
NG tubes¶	Various sized
60 cc irrigation syringe¶	2

Monitoring equipment	
Esophageal or other core temperature probes (eg, for nasopharyngeal, tympanic membrane, rectal, or bladder sites, or a pulmonary artery catheter)¶	
CVP kits (sizes appropriate to your patient population)¶	Sizes appropriate for patient population
Transducer kits for arterial and central venous cannulation¶	
Nursing supplies	
Large sterile plastic drape (eg, Steri-Drape, for rapid drape of wound)¶	1
Urine meter¶	1
Irrigation tray with piston (60 cc irrigation) syringe¶	1
Small and large clear plastic bags for ice¶	4 each
Bucket for ice¶	1
Test strips: urine analysis for hemoglobin and finger stick glucose¶	1 vial
Laboratory testing supplies	
Syringes (3 mL) for blood gas analysis or ABG kits × 6 or supplies for point-of-care testing device (eg, iSTAT).	
Blood specimen tubes for CK, myoglobin, CMP, thyroid studies, PT/PTT, fibrinogen, fibrin split products, lactate, and CBC. If no immediate laboratory analysis is available, samples should be kept on ice for later analysis. This may well prove useful on retrospective review and diagnosis. Blood cultures are very useful and should be included to rule out bacteremia.	
Urine collection container for myoglobin level. Pigmenturia (eg, brown or red urine and heme-positive dipstick) indicates that renal protection is mandated, when the urine is centrifuged or allowed to settle, and the sample shows clear supernatant, ie, the coloration is due to red cells in the sample.	

MH: malignant hyperthermia; IV: intravenous; NG: nasogastric; G: gauge; CVP: central venous pressure; ABG: arterial blood gas; CK: creatine kinase; CMP: comprehensive metabolic panel; PT: prothrombin time; PTT: partial thromboplastin time; CBC: complete blood count.

* We suggest that institutions establish the number and location of MH treatment carts so that one or more carts is available immediately or within no more than 10 minutes to all anesthetizing locations in the facility including operating rooms and procedural suites.

¶ These supplies are only necessary if not already immediately accessible at all anesthetizing locations in the institution.

Adapted from: What Should be on an MH Cart? The Malignant Hyperthermia Association of the United States. Available at:

https://www.mhaus.org/healthcare-professionals/be-prepared/what-should-be-on-an-mh-cart/ (Accessed August 9, 2018).

Graphic 67323 Version 14.0

Table 2: **Sequence of clinical signs of malignant hyperthermia**

Frequent initial clinical signs of MH
▪ Hypercarbia (elevated end-tidal CO_2) resistant to increases in minute ventilation
▪ Tachypnea (or breathing over the ventilator)
▪ Sinus tachycardia
Unusual initial clinical signs of MH
▪ Masseter spasm with or without administration of succinylcholine
▪ Generalized muscular rigidity
▪ Peaked T waves or other arrhythmia (including ventricular tachycardia or fibrillation) resulting from sudden acute hyperkalemia
▪ Hyperthermia (variable rates of rise)
▪ Sweating
Delayed clinical sequelae of MH
▪ Cola-colored urine (myoglobinuria) resulting from significant rhabdomyolysis
▪ Disseminated intravascular coagulation (from prolonged or severe hyperthermia)

The sequence and timing of clinical presentation of MH varies. This table shows the initial and delayed signs of MH, based on reported cases.

MH: malignant hyperthermia; CO_2: carbon dioxide.

Graphic 119718 Version 1.0

Table 3: **Prevalence of clinical signs during malignant hyperthermia episode**

Clinical sign	Percentage with sign
Hypercarbia	92.2
Sinus tachycardia	72.9
Rapidly increasing temperature	64.7
Elevated temperature	52.2
Generalized muscular rigidity	40.8
Tachypnea	27.1
Masseter spasm	26.7
Sweating	17.6
Cola-colored urine	13.7
Cyanosis	9.4
Ventricular tachycardia	3.5
Excessive bleeding	2.7
Ventricular fibrillation	2.4

Clinical signs of malignant hyperthermia, listed with the prevalence with which each was noted in 255 patients reported to the North American MH Registry from 1987 to 2006. Signs such as hypercarbia and tachycardia were defined as being inappropriate to the clinical situation. Notably, 79.6 percent had one or more signs indicative of muscular involvement (masseter spasm, generalized muscular rigidity, cola-colored urine, peak creatinine kinase >10,000 units/L, or peak potassium >6.0 mEq/L).

Data from: Larach MG, Gronert GA, Allen GC, et al. Clinical presentation, treatment, and complications of malignant hyperthermia in North America from 1987 to 2006. Anesth Analg 2010; 110:498.

Graphic 88906 Version 3.0

Table 4: **Laboratory values in acute malignant hyperthermia**

Laboratory study	Laboratory value that supports the diagnosis of MH
$PaCO_2$	>60 mmHg during controlled ventilation
	>65 mmHg during spontaneous ventilation
Serum potassium	>6 mEq/L, in patients without renal failure
Arterial pH	<7.25
Base deficit	≥8 mEq/L
Creatine kinase*	>20,000 units/L after administration of succinylcholine
	>10,000 units/L without administration of succinylcholine
Urine myoglobin*	>60 mcg/L
Serum myoglobin*	>170 mcg/L

Typical laboratory values used to confirm the diagnosis of acute malignant hyperthermia.

$PaCO_2$: partial pressure of arterial carbon dioxide; MH: malignant hyperthermia.
* Creatine kinase and myoglobin levels peak at approximately 14 hours after an acute MH event.

Adapted from: Larach MG, Localio AR, Allen GC, et al. A clinical grading scale to predict malignant hyperthermia susceptibility. Anesthesiology 1994; 80:771.

Graphic 88983 Version 5.0

Table 5: **Physical findings and laboratory values for pediatric malignant hyperthermia**

Characteristic, N	Youngest (n = 35)		Middle (n = 163)		Oldest (n = 66)		
Physical exam findings	**n**	**%**	**n**	**%**	**n**	**%**	**P**
Elevated temperature	21	60.0	71	43.6	36	54.5	0.1
Rapidly increasing temperature	13	37.1	51	31.3	44	66.7	<0.001
Sweating	0	0.0	11	6.7	17	25.8	<0.001
Masseter spasm	5	14.3	74	45.4	7	10.6	<0.001
Generalized muscular rigidity	6	17.1	53	32.5	25	37.9	0.96
Dark urine	2	5.7	24	14.7	6	9.1	0.26
Tachypnea	12	34.3	43	26.4	21	31.8	0.52
Hypercarbia	27	77.1	95	58.3	59	89.4	<0.001
Cyanosis	2	5.7	16	9.8	7	10.6	0.75
Skin mottling	5	14.3	8	4.9	7	10.6	0.07
Sinus tachycardia	26	74.3	110	67.5	57	86.4	0.01
Ventricular tachycardia	2	5.7	9	5.5	3	4.5	1
Ventricular fibrillation	0	0.0	5	3.1	4	6.1	0.27
Vital sign and laboratory data	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**P**
Maximum temperature (°C)	38.4	(1.4)	38.2	(1.7)	40.2	(8.3)	0.0006
Time to maximum temperature (minutes)	181.7	(282.3)	137.1	(191.3)	146.9	(183.0)	0.07
Maximum end-tidal CO_2 (mmHg)	75.9	(23.0)	62.8	(23.9)	75.2	(16.8)	0.0001
Time to maximum $etCO_2$ (minutes)	640.8	(205.6)	682.1	(228.9)	861.7	(329.7)	0.0001
Time to first blood gas (minutes)	183.6	(253.5)	151.2	(219.3)	176.2	(212.3)	0.18
Fio_2	0.75	(0.35)	0.78	(0.32)	0.82	(0.28)	0.51

pH	7.16	(0.19)	7.23	(0.14)	7.2	(0.13)	0.06
Pco_2 (mmHg)	65.3	(39.4)	53.2	(24.1)	61.6	(18.9)	0.0008
Po_2 (mmHg)	157.6	(142.4)	236.5	(176.1)	275.5	(162.2)	0.002
Base excess (mEq/L)	-6.6	(5.4)	-5.6	(6.2)	-4.9	(5.7)	0.19
Bicarbonate (mEq/L)	21.69	(4.58)	21.34	(4.91)	22.75	(3.98)	0.052
Peak lactic acid (mmol/L)	5.2	(5.8)	1.7	(3.4)	3.8	(2.3)	0.003
Peak potassium (mEq/L)	5.07	(1.30)	5	(1.68)	5.44	(1.24)	0.004
Peak creatinine kinase (units/L)	17,843.5	(58,940.6)	32,895.8	(70,279.5)	20,527.8	(61,000.7)	0.0001

$etCO_2$: end-tidal carbon dioxide.

From: Nelson P, Litman RS. Malignant Hyperthermia in Children: An Analysis of the North American Malignant Hyperthermia Registry. Anesth Analg 2014; 118:369. DOI: 10.1213/ANE.0b013e3182a8fad0. Reproduced with permission from Lippincott Williams & Wilkins. Copyright © 2014 International Anesthesia Research Society. Unauthorized reproduction of this material is prohibited.

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Table 6: **Clinical indicators of acute malignant hyperthermia**

Hypercarbia*	Controlled ventilation:
	$ETCO_2$ >55 mmHg
	$PaCO_2$ >60 mmHg
	Spontaneous ventilation:
	$ETCO_2$ >60 mmHg
	$PaCO_2$ >65 mmHg
Muscle rigidity	Generalized muscular rigidity (sustained contracture) during anesthesia with triggering agents
	Severe masseter muscle spasm shortly following succinylcholine administration that does not abate within 1 minute
Arrhythmias	Tachycardia
	Ventricular arrhythmias (indicating acute hyperkalemia)
Acidosis	Primarily respiratory acidosis but may include varying degrees of metabolic acidosis
	Arterial pH <7.25
	Base deficit >8 mEq/L
Hyperthermia	Core temperature >38.8°C
	Rapid rise in temperature
Rhabdomyolysis	CK >20,000 international units after anesthesia with succinylcholine
	CK >10,000 international units after anesthesia without succinylcholine
	Cola-colored urine in perioperative period
	Urine myoglobin >60 mcg/L
	Serum myoglobin >170 mcg/L
	Serum potassium >6 meq/L (without renal failure)

$ETCO_2$: end-tidal carbon dioxide; $PaCO_2$: partial pressure of arterial carbon dioxide; CK: creatine kinase.
* These values are approximate guidelines determined by experts with knowledge of and expertise in managing malignant hyperthermia events.

Graphic 54539 Version 7.0

Table 7: **Malignant hyperthermia rapid overview of emergency management**

Clinical signs*
Hypercarbia (elevated $ETCO_2$) resistant to increases in minute ventilation
Tachypnea or breathing over the ventilator
Sinus tachycardia
Masseter spasm with or without administration of succinylcholine
Generalized muscle rigidity
Peaked T waves or other arrhythmia (including PVCs, ventricular tachycardia or fibrillation) as a result of hyperkalemia
Mixed acidosis on blood gas
Hyperthermia
Sweating
Management
Call for help and MH cart; for questions at any time call MH hotline: 1-800-644-9737 in US, 1-209-417-3722 outside US
Discontinue inhaled anesthetics and succinylcholine; increase fresh gas flow to ≥10 L/minute, use non-triggering agents for remainder of procedure
Notify surgeon; complete surgical procedure as quickly as possible
Hyperventilate with 100% oxygen, perform endotracheal intubation if ETT not in place
Insert carbon filters into breathing circuit after flushing the breathing circuit for ≥90 seconds at ≥10 L/minute fresh gas flow
Administer dantrolene:
▪ Initial dose 2.5 mg/kg IV rapid bolus; eg, for a 70 kg patient, administer 175 mg IV (9 vials of diluted dantrolene or 3.5 mL of diluted Ryanodex)
• For older dantrolene preparations (ie, Dantrium, Renovo, or dantrolene sodium), dilute each 20 mg vial with 60 mL sterile preservative-free water; eg, for 70 kg patient, prepare nine 20 mg vials
• For Ryanodex, dilute a 250 mg vial with 5 mL sterile preservative-free water
▪ Watch for reversal of clinical signs ($ETCO_2$ should begin to normalize); repeat dantrolene bolus (2.5 mg/kg IV) as necessary; cumulative doses ≥10 mg/kg IV may be required
Send laboratory studies: venous or arterial blood gases, electrolytes, CK; repeat as necessary
▪ Treat hyperkalemia in patients with arrhythmias or potassium >6 mEq/L
• Calcium:
◦ Calcium chloride

▪ Adult: 0.5 to 1 g IV (5 to 10 mL of 10% solution) per dose;
▪ Pediatric: 10 to 20 mg/kg IV (0.1 to 0.2 mL/kg 10% solution), maximum 2 g (20 mL) per dose
Repeat after five minutes if ECG changes persist
or
○ Calcium gluconate
▪ Adult: 1.5 to 3 g IV (15 to 30 mL of 10% solution);
▪ Pediatric: 60 to 100 mg/kg IV (0.6 to 1 mL/kg of 10% solution), maximum 3 g (30 mL) per dose
• Sodium bicarbonate: 1 to 2 mEq/kg IV push over 5 to 10 minutes (maximum 100 mEq per dose); do not administer sodium bicarbonate in the same line as calcium
• Insulin and dextrose: check blood glucose hourly
○ Adult: 10 units IV regular insulin IV push with 50 mL IV 50% dextrose
○ Pediatric: 0.1 units/kg insulin IV push with 0.5 g/kg dextrose (eg, 1 mL/kg 50% dextrose or 2 mL/kg 25% dextrose)
▪ Treat metabolic acidosis with base deficit ≥8 mEq/L with sodium bicarbonate 1 to 2 mEq/kg IV over 5 to 10 minutes, maximum 100 mEq per dose.
Treat arrhythmias per ACLS, avoid calcium channel blockers; most arrhythmias respond to correction of hyperkalemia and acidosis
Cool the patient as necessary: Start cooling for core temperature >39°C, discontinue cooling when temperature decreases to 38°C
Insert Foley catheter, maintain urine output at 1 to 2 mL/kg/hour with IV fluid and diuretics
Ongoing care
Arrange ICU bed for at least 24 hours; monitor for recurrence, rhabdomyolysis, DIC
After initial MH event is controlled, administer dantrolene 1 mg/kg IV every four to six hours or 0.25 mg/kg/hour for at least 24 hours

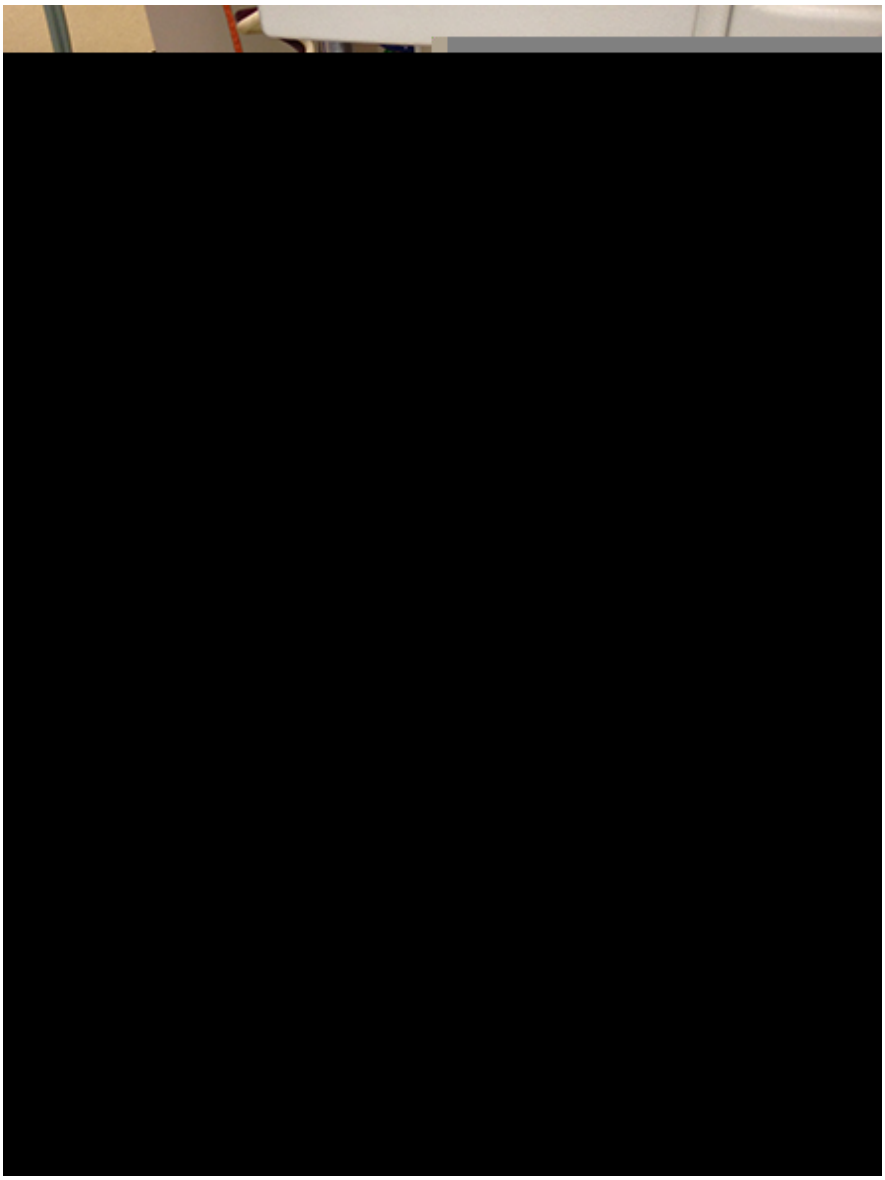
For further information, refer to UpToDate content on management of an acute MH crisis.

ACLS: advanced cardiac life support; CK: creatine kinase; DIC: disseminated intravascular coagulation; ECG: electrocardiogram; $ETCO_2$: end-tidal carbon dioxide; ETT: endotracheal tube; ICU: intensive care unit; IV: intravenous; MH: malignant hyperthermia; PVCs: premature ventricular contractions.

* The sequence and timing of the clinical manifestations of MH vary from patient to patient, and most patients do not develop all signs of MH.

Graphic 68918 Version 15.0

Picture 1: **Anesthesia machine charcoal filter**



Anesthesia machines no longer need to be flushed through with oxygen prior to use by an MHS patient. A charcoal filter attached to the inspiratory and expiratory breathing circuit limbs effectively prevents residual concentrations of anesthetic gases from reaching the patient. It should also be used when an acute MH crisis is diagnosed to limit further exposure of the patient to anesthetic gases.

MHS: susceptibility to malignant hyperthermia; MH: malignant hyperthermia.

Graphic 81599 Version 9.0

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